# Disposal of wastewater and oxygen sag curve

Module 3 Lecture 6

# Disposal by dilution

- In this process, the raw sewage or the partially treated sewage is thrown into natural waters having large volume.
- The sewage in due course of time is purified by what is known as the self-purification capacity of natural waters. The limit of discharge and degree of treatment of sewage are determined by the capacity of self-purification of natural waters.
- Natural waters to which sewage can be disposed include
    1. Creeks
    2. Estuaries
    3. Ground waters
    4. Lakes
    5. Ocean or sea
    6. Perennial rivers and streams

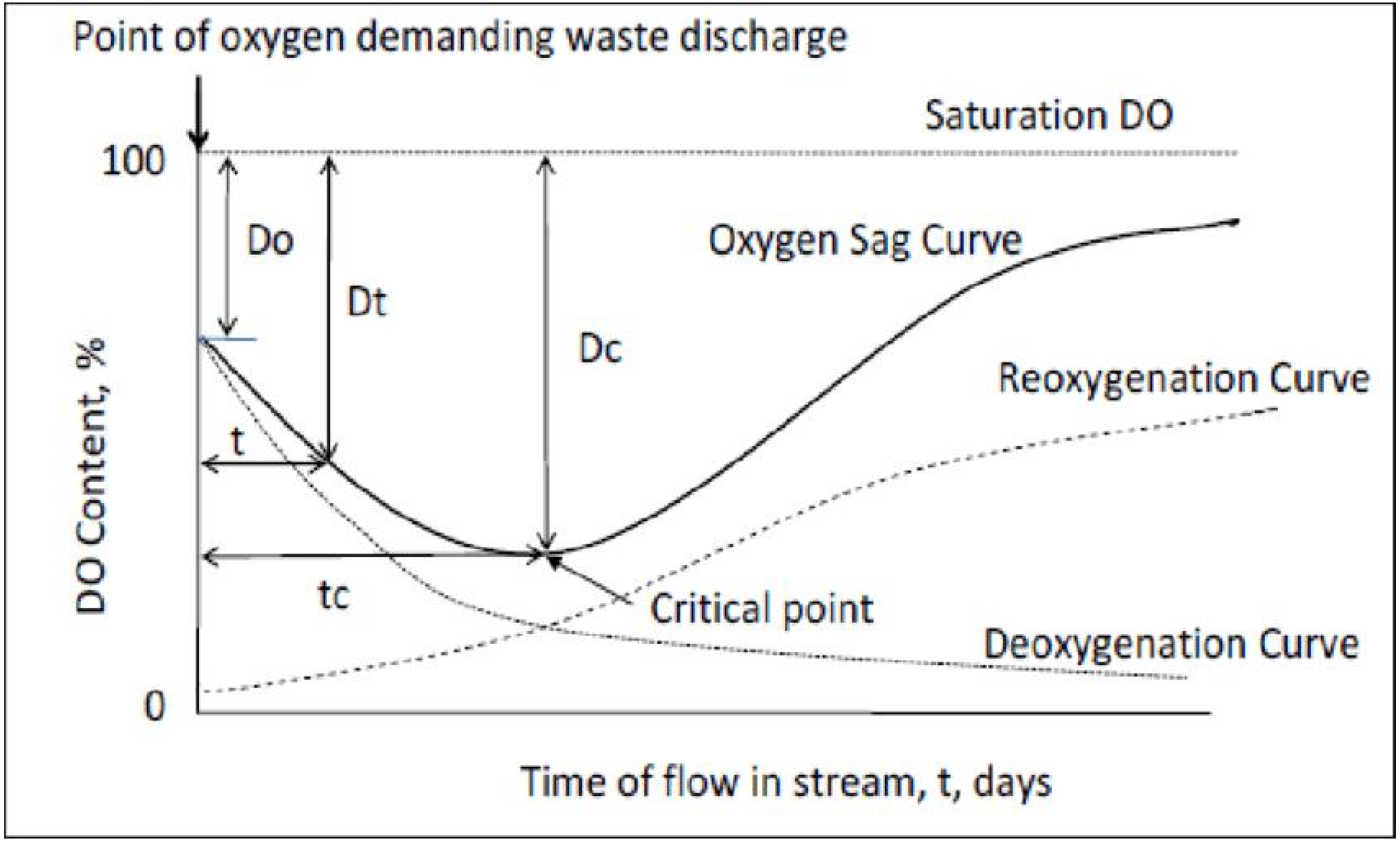
# Conditions favorable for dilution

- Currents of flow of diluting waters should be favourable which means that nuisance should not be caused when sewage is discharged into diluting waters
- Diluting waters are not used for the purpose of navigation for at least some reasonable distance on the downstream from the point of sewage disposal.
- Diluting waters should not have habitation or they should not have been used as source of water supply for at least some reasonable distance on the downstream from the point of sewage disposal
- Dissolved oxygen content of diluting waters should be high
- The place is situated near natural waters having large volumes
- The sewage is relatively fresh and it is possible to bring it to the point of discharge within four or five hours of its production.

# Zones of pollution in a river or Stream

| | Zones of pollution | | | | |
|---|---|---|---|---|---|
| | Clear water | Zone of degridation | Zone of active decomposition | Zone of recovery | Zone of clearer water |
| Dissolved oxygen sag curve | Saturation level 100% D.O. 40% Zero% | 40% | | | |
| Physical indices | Clear water, no bottom sludge, no colour | Floating solids; bottom sludge present, colour getting turbid | Darker and greyish colour, evolution of gases like $CH_4$,$CO_2$,$H_2S$ etc. lot of sludge coming to the surface forming an ugly scum layer at top | Turbid with bottom sludge | Clear water with no bottom sludge |
| Fish presence | Ordinary fish like game, pan, food & for age etc. present | Tolerant fishes like carps, buffalo, gary, etc. present | No fish present | Tolerant fish like carp, buff alo, etc. are present | Ordinary fish like game, pan, food, and forage, etc. present |
| Bottom Animals | | | | | |
| Algae & Protozoa etc called plankton | | | | | |

# Oxygen sag curve




Oxygen deficit, $D_t$ = $DO_s$- $DO_A$

$D_o$ = initial oxygen deficit

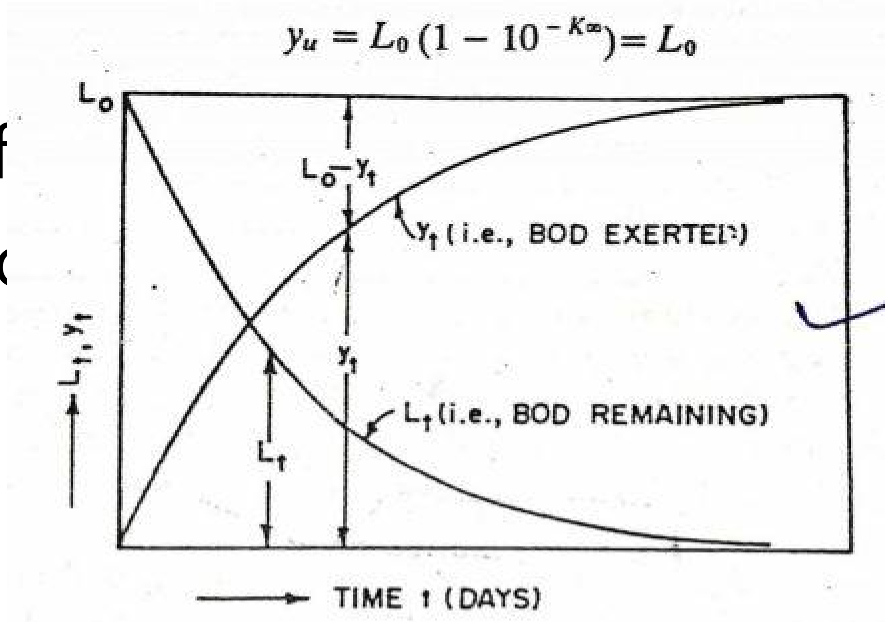
$D_c$ = critical deficit

Deoxygenation – f($L_0$, T)

Reoxygenation = f(DO deficit, T, Vel, Depth)

# First stage BOD

- Assumption : At a given time rate of deoxygenation is assumed to be direc proportional to the rate of organic matter present in sewage.




- $\frac{dL_t}{dt} = -kL_t$
- $L_t$ – first stage BOD remaining at any time t in mg/L
- K – rate constant (rate of oxidation – depends on nature of organic matter and temperature) in day$^{-1}$

# First stage BOD

- $\frac{dL_t}{dt} = -kL_t$

Integrating and applying boundary condition as $L_t = L_0$ at t = 0 days

- $log_e \frac{L_t}{L_0} = -kt$
- $\frac{L_t}{L_0} = e^{-kt} = 10^{-k't}$ $OR\ L_t = L_0 \times 10^{-k't}$
- where k′ is the deoxygenation constant ( 0.05 – 0.3 $day^{-1}$)
- Amount of BOD exerted, $y_t = (L_0 - L_t) = L_0 (1 - 10^{-k't})$
- $k_t = k_{20}\theta^{(T-20°)}$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The $BOD_5$ of a waste has been measured as 600 mg/L. If the rate constant k = 0.23/day (base e), what is the Ultimate BOD of the waste, what proportion of $BOD_u$ will remain unoxidized after 20 days. ($K_1$=K*0.434)

a) $Y_t = L(1-10^{-k_D t})$

$k = 0.23/\text{day}$

$k_D = 0.434\,k = 0.434 \times 0.23 = 0.1$

$Y_5 = 600\,mg/l = L\left[1-10^{-(0.1\times 5)}\right]$

$600 = L\left[1-(10)^{-0.5}\right]$

$= L\left[1-\frac{1}{3.16}\right] = L\left[1-0.316\right]$

$= 0.684\,L$

$L = \frac{600}{0.684} = 877.5\,mg/l$

$BOD_u = \underline{\underline{877.5}}\,mg/l$

b) $Y_{20} = L(1-10^{(-0.1\times 20)})$

$= Y_u\left[1-0.01\right] = Y_u(0.99)$

$Y_{20} = 0.99\,Y_u$

It means that 99% of $BOD_u$ is utilised in 20 days.

The BOD of a sewage incubated for one day at 30˚C has been found to be 100 mg/L. what will be the 5 day 20˚C BOD. Assume k′ = 0.12 (base 10) at 20˚C

a) $Y_1 = 100$ mg/li

$k_{20} = 0.12$

Find $k_{30}$

$$K_T = K_{20}\,\theta^{T-20} \quad \text{——— ①}$$

The temperature range is 20° to 30°C,

$\theta = 1.056$

$$k_{30} = k_{20}(1.056)^{30-20}$$

$$= 0.207/\text{day}.$$

$$y_{t\,(30°)} = L_0\left(1-10^{-k_{30}t}\right) \quad \text{——— ②}$$

$$100 = L_0\left(1-10^{-0.207\times t}\right)$$

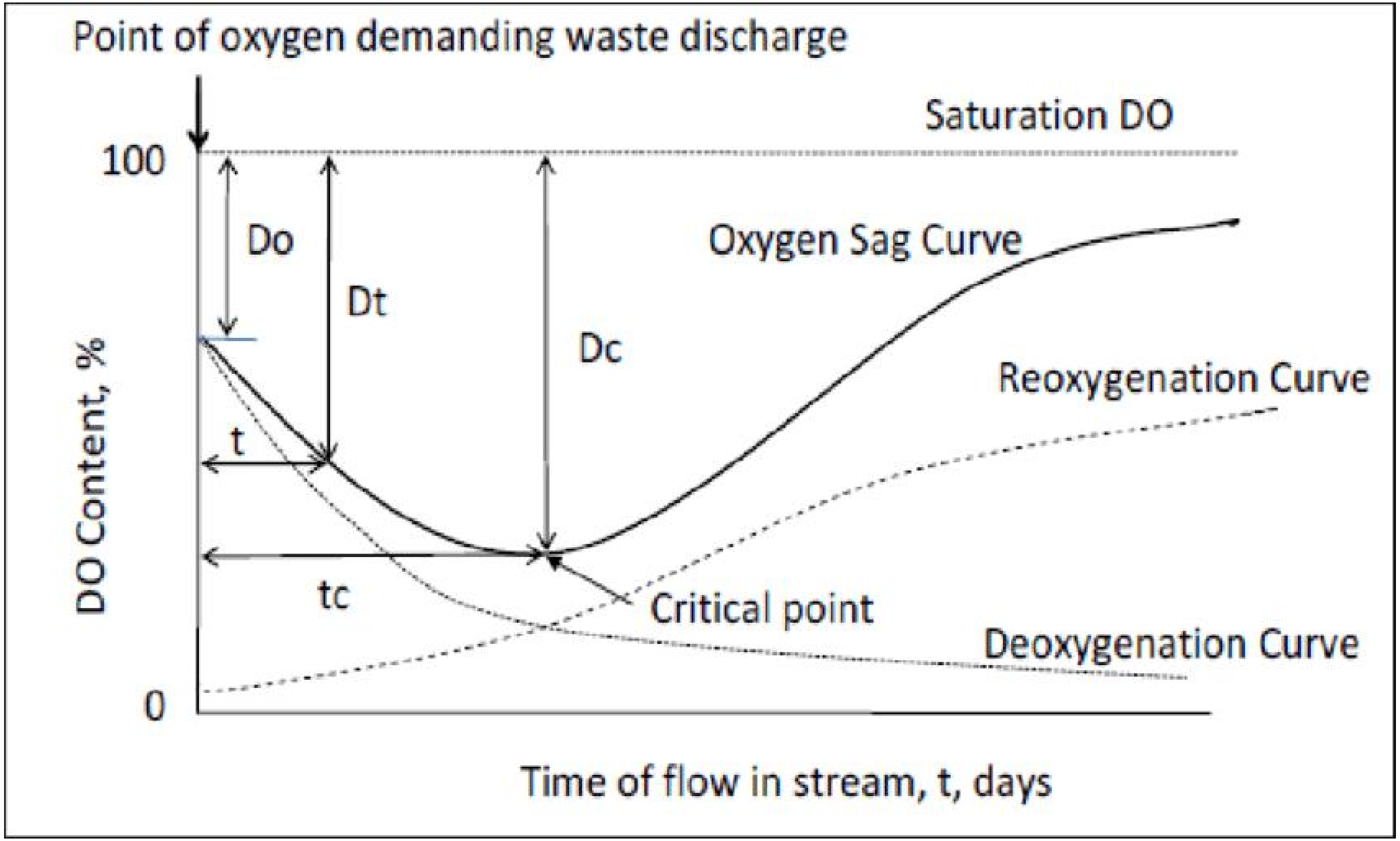
$$L_0 = \underline{\underline{263.8}}\ \text{mg/l}$$

$$y_5(20°) = L_0\left(1-10^{-k_{20}t}\right)$$

$$= 263.8\left(1-10^{-0.12\times 5}\right)$$

$$= 197.5$$

$BOD_5$ at 20°C = $\underline{\underline{197.5}}$ mg/l.

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$D_c$ = critical deficit

Deoxygenation – f($L_0$, T)
Reoxygenation = f(DO deficit, T, Vel, Depth)

# Streeter-Phelps Equation

$$D_t = \frac{k' L_0}{R' - k'}\left(10^{-k't} - 10^{-R't}\right) + D_0 10^{-R't}$$

$$D_c = \frac{k'}{R'} L_0 10^{-k't_c}$$

$$t_c = \frac{1}{R' - k'} log_{10} \frac{R'}{k'}\left[1 - \frac{D_0 (R' - k')}{k' L_0}\right]$$

- $x_c = t_c \times velocity\ of\ flow$
- $\frac{R}{k} = \frac{R'}{k'} = f,\ self\ purification\ constant$
- $R_T = R_{20} 1.024^{(T-20)}$

- $k_T = k_{20} \theta^{(T-20)}$

- A stream saturated with DO has a flow of 1.2 $m^3/s$, BOD of 4 mg/L and reoxygenation rate constant (base 10) of 0.3 per day. It receives an effluent discharge of 0.25 $m^3/s$ having BOD 20 mg/L, DO – 5 mg/L and deoxygenation rate constant (base 10) of 0.13/day. The average velocity of flow of the stream is 0.18 m/s. calculate the DO deficit at a point 20 km and 40 km downstream. Assume that the temperature is 20˚C throughout and BOD is measured at 5 days. Take saturation DO as 9.17 mg/L

a) BOD of effluent = 20 mg/l

BOD of stream = 4 mg/l

$$\text{BOD of mix} = Y_m = \frac{Q_s Y_s + Q_e Y_e}{Q_s + Q_e}$$

$Q_s = 1.2 m^3/s$

$Q_e = 0.25 m^3/s$

$$= \frac{1.2 \times 4 + 0.25 \times 20}{1.2 + 0.25}$$

$$= \underline{\underline{6.759}} \text{ mg/l}$$

$$Y_5 = Y_{mix} = L_0 (1 - 10^{-k't})$$

$$6.759 = L_0 (1 - 10^{-0.13 \times 5})$$

$k = 0.13/\text{day}$

$$L_0 = \underline{\underline{8.71}} \text{ mg/l}$$

$DO_s = 9.17$ mg/l

$DO_e = 5$ mg/l

$$(DO)_{mix} = \frac{(DO)_s (Q_s) + (DO)_e (Q_e)}{Q_s + Q_e}$$

$$= \frac{9.17 \times 1.2 + 5 \times 0.25}{1.2 + 0.25}$$

$$= \underline{\underline{8.45}} \text{ mg/l}$$

Initial DO deficit $D_0$ = $9.17 - 8.45 = \underline{0.72}$ mg/l

a) D.O deficit at a point 20 km D/stream,

$$t = \frac{\text{distance}}{\text{velocity}} = \frac{20 \times 1000}{0.18 \times 60 \times 60 \times 24} = \underline{\underline{1.286 \text{ days}}}$$

Using Streeter Phelps equation:

$$D_t = \frac{k' L_0}{R' - k'}\left(10^{-k't} - 10^{-R't}\right) + D_0 \, 10^{-R't}$$

$$= \frac{0.13 \times 8.71}{0.3 - 0.13}\left(10^{-0.13 \times 1.286} - 10^{-0.3 \times 1.286}\right) + 0.72 \times 10^{-0.3 \times 1.286}$$

$$= \underline{\underline{2.089 \text{ mg/l}}}$$

b) D.O deficit at a point 40 km downstream,

$$t = \frac{40 \times 1000}{0.18 \times 60 \times 24 \times 60} = 2.572 \text{ days}$$

$$D_t = \frac{0.13 \times 8.71}{0.3 - 0.13}\left(10^{-0.13 \times 2.572} - 10^{-0.3 \times 2.572}\right) + 0.72 \times 10^{-0.3 \times 2.572}$$

$$= \underline{\underline{2.079 \text{ mg/l}}}$$

- A town discharges 80 cumecs of sewage into a stream having a rate of flow of 1200 cumecs during lean days, and the five day BOD of the sewage at the given temperature is 250 mg/L. Find the amount of critical DO deficit and the initial BOD . Assume K’ as 0.1, assume saturation DO at given temperature as 9.2 mg/L

$P_n$)

$(D.O)_{stream} = 9.2$ mg/l $\quad (DO)_{eff} = 0$

$$(DO)_{mix} = \frac{(9.2 \times 1200) + (0 \times 80)}{1200 + 80} = \underline{\underline{8.625}} \text{ mg/l}$$

Initial DO deficit $= D_0 = 9.2 - 8.625$

$= \underline{\underline{0.575}}$ mg/l

$$Y_5 = (BOD)_{mix} = \frac{(0 \times 1200) + (250 \times 80)}{1200 + 80} = 15.625 \text{ mg/l}$$

$$Y_5 = L_0 (1 - 10^{-k't})$$

$$15.625 = L_0 (1 - 10^{-0.1 \times 5})$$

$$L_0 = \underline{\underline{22.85}} \text{ mg/l}$$

# Disposal by land treatment

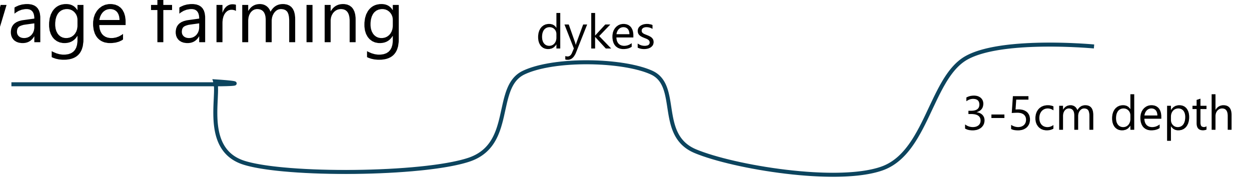
- The raw domestic wastewater (sewage) is applied on the land.
- A part of sewage evaporates and the remaining portion percolates through the ground and is caught by the underground drains for disposal into natural waters.
- The sewage adds to the fertilizing value of land and crops can be profitably raised on such land.
- The term sewage farming is also sometimes used for indicating disposal of sewage by land treatment.

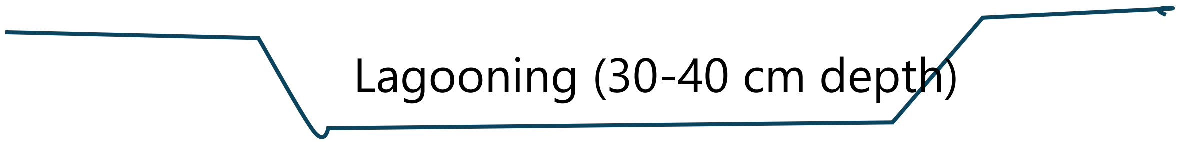
# Conditions favourable for land treatment

- The area of land treatment is composed of sandy, loamy or alluvial soils.  Such soils are easily aerated, and it is easy to maintain aerobic conditions in them
- The depth of water table is more even in rainy season so that there are no chances of pollution of underground water sources by land treatment
- The rainfall in the area is low as it will assist in maintaining good absorption capacity of soil
- There is absence of river or other natural water sources in the vicinity of disposal of sewage
- There is demand for cash crops which can be easily grown on sewage farms
- There is availability of large open areas in the surrounding locality for practicing broad irrigation by sewage

# Land treatment

- Broad irrigation or sewage farming
  - Flooding
  - Surface irrigation
  - Spray irrigation
  - Ridge and Furrow
    - Straight run
    - Zig-zag
  - Subsurface irrigation –rootzone irrigation
- Rapid infiltration
- Overland runoff



# Sewage sickness

- If sewage is applied continuously on a piece of land, pores or voids of soil are filled up or clogged.
- Free circulation of air is thereby foul-smelling and anaerobic conditions develop.
- At this stage, the land is unable to take any further sewage load.
- Organic matter decomposes and foul-smelling gases are produced.
- The phenomena of soil is known as sewage sickness of land.

# Sewage sickness

- Primary Treatment of sewage
- Choice of Land
- Under Drainage of soil
- Giving rest to land
- Rotation of crops
- Applying shallow depth

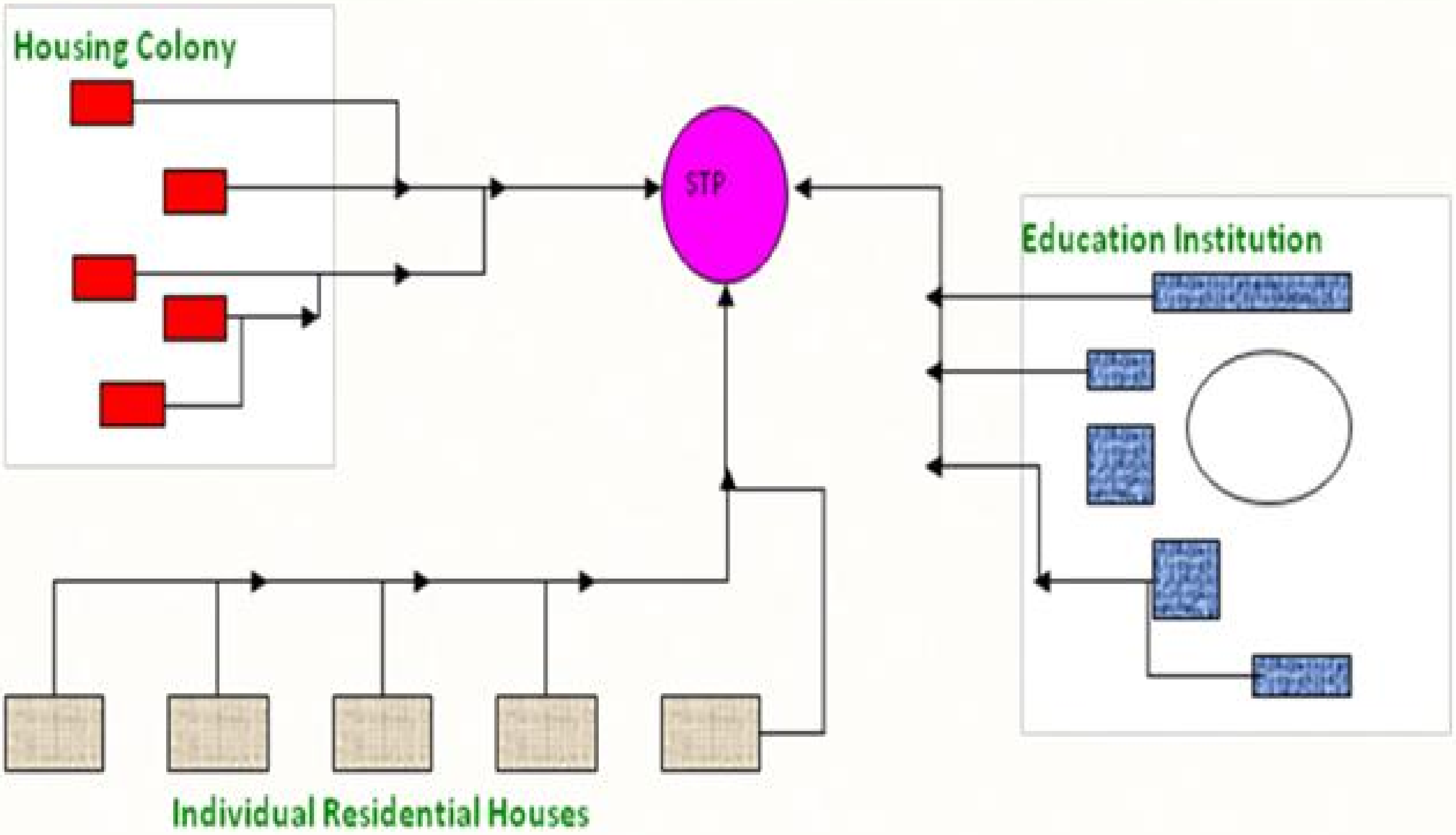
# Decentralised and Natural methods of wastewater treatment

Module 3

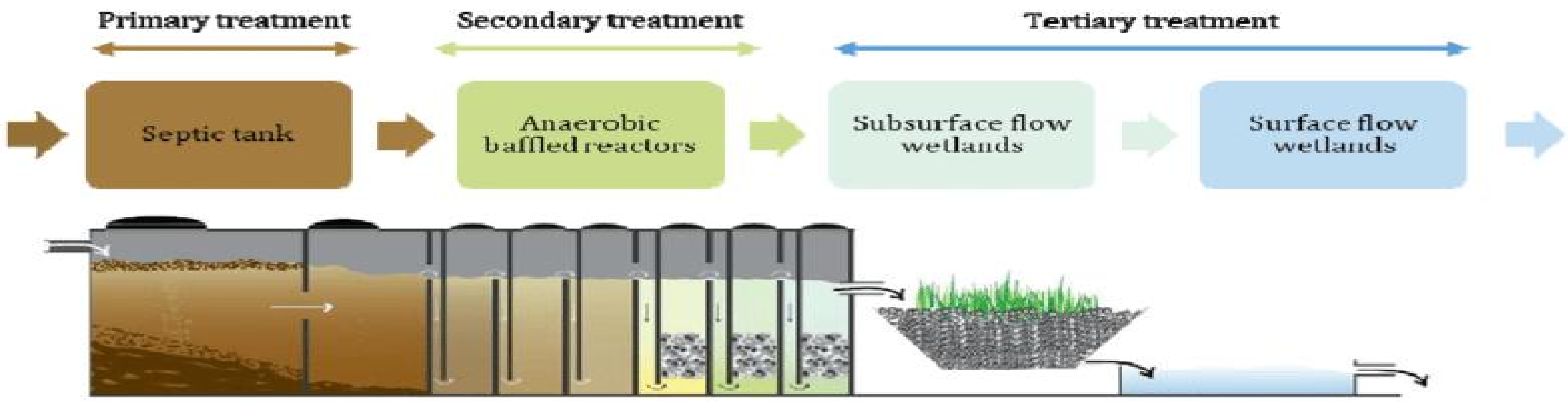
# DECENTRALIZED WASTEWATER TREATMENT

- Decentralized wastewater management (DWWM) may be defined as “the collection, treatment, and disposal/reuse of wastewater from individual homes, clusters of homes, isolated communities, industries, or institutional facilities, as well as from portions of existing communities at or near the point of waste generation”

- In case of decentralized systems, both solid and liquid fractions of the wastewater are utilized near their point of origin, except in some cases when a portion of liquid and residual solids may be transported to a centralized point for further treatment and reuse.

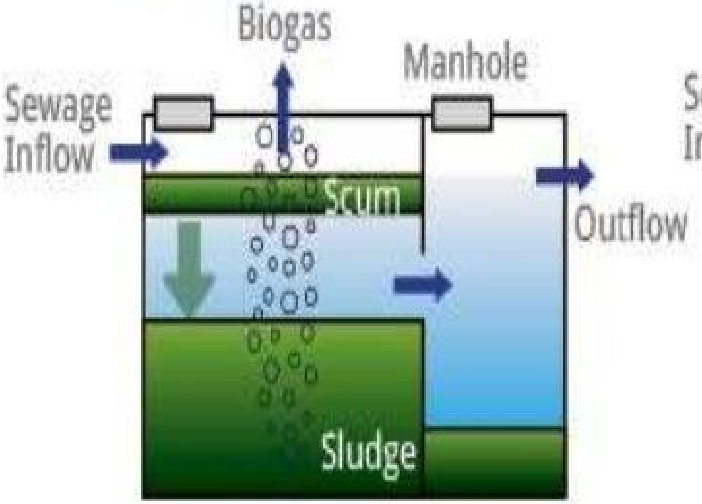



Schematic Network Diagram of Decentralized Wastewater Treatment System

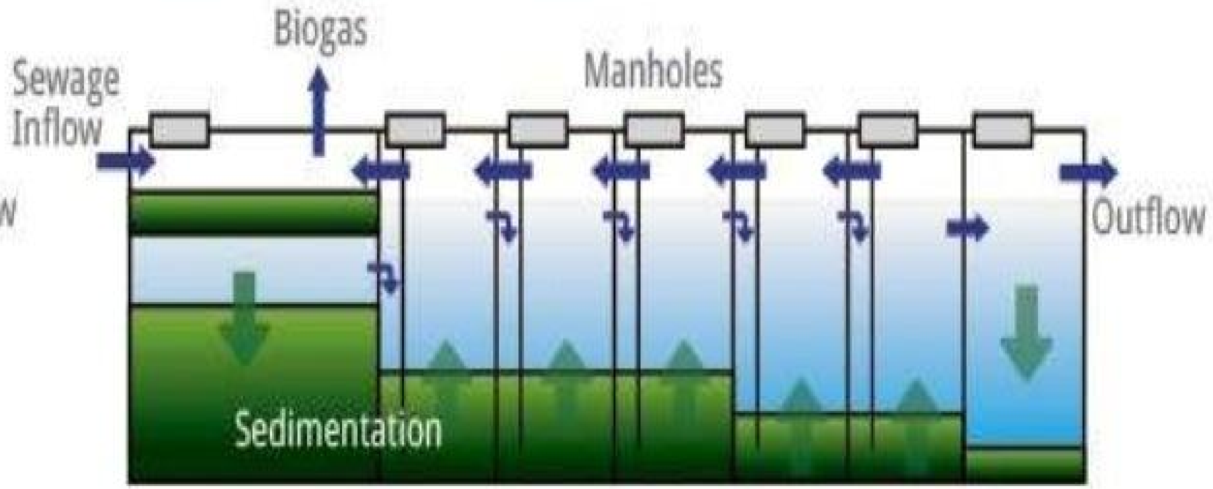



| | Section 1 | Section 2 | Section 3 | Section 4 | Section 5 |
|---|---|---|---|---|---|
| | Primary treatment | Secondary treatment | | Tertiary treatment | Post-treatment |
| Methods | - Septic Tank<br>- Cesspools<br>- Holding Tanks<br>- Ecological Sanitation | - Waste Stabilization Ponds<br>- Media Filters<br>- Membrane Biological Reactors<br>- Anaerobic Digestion<br>- Terra Preta Sanitation | | - Constructed wetland<br>- Sand filters<br>- Peat filters<br>- Vermifilters<br>- Wastewater stabilization ponds | - Polishing pond/ Surface flow wetland |
| | Sedimentation tank stabilizing settled sludge by anaerobic digestion | Baffled reactors - Anaerobic degradation of suspended and dissolved solids | Anaerobic filter - Water passes through filter media | Open shallow basin filled with gravel/pebbles to support growth of plant/reeds with shallow roots | Open shallow basin for removal of stabilized or inactive suspended substances |

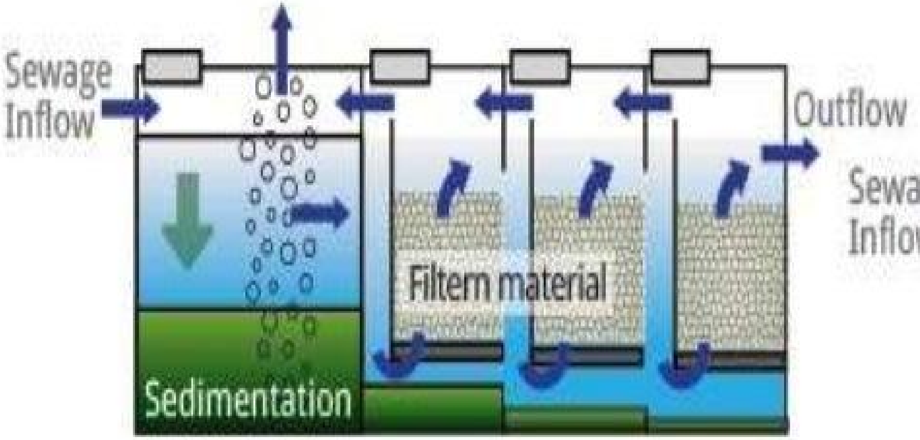
## 1. Settler



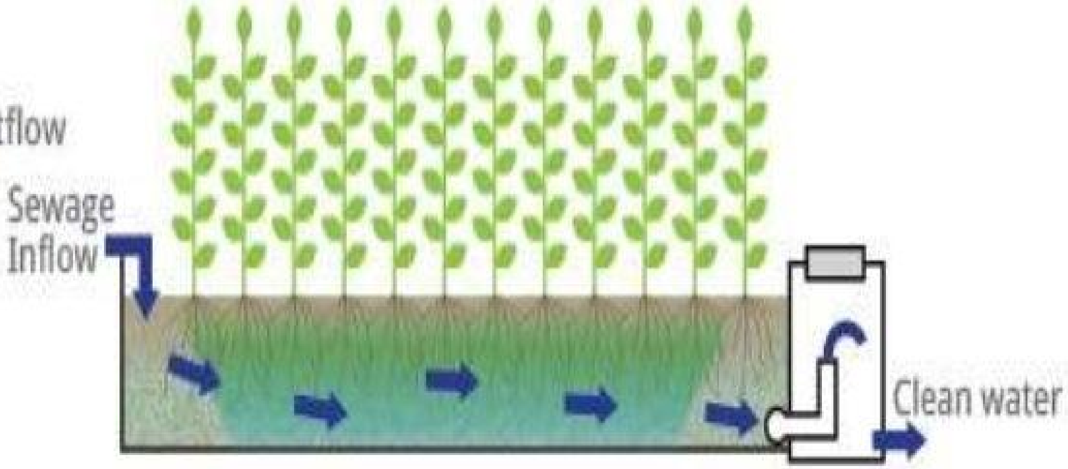

## 2. Anaerobic Baffled Reactor




## 3. Anaerobic Filter




## 4. Planted Gravel Filter



# ADVANTAGES OF DECENTRALIZED WASTEWATER MANAGEMENT SYSTEMS (DWWMS)

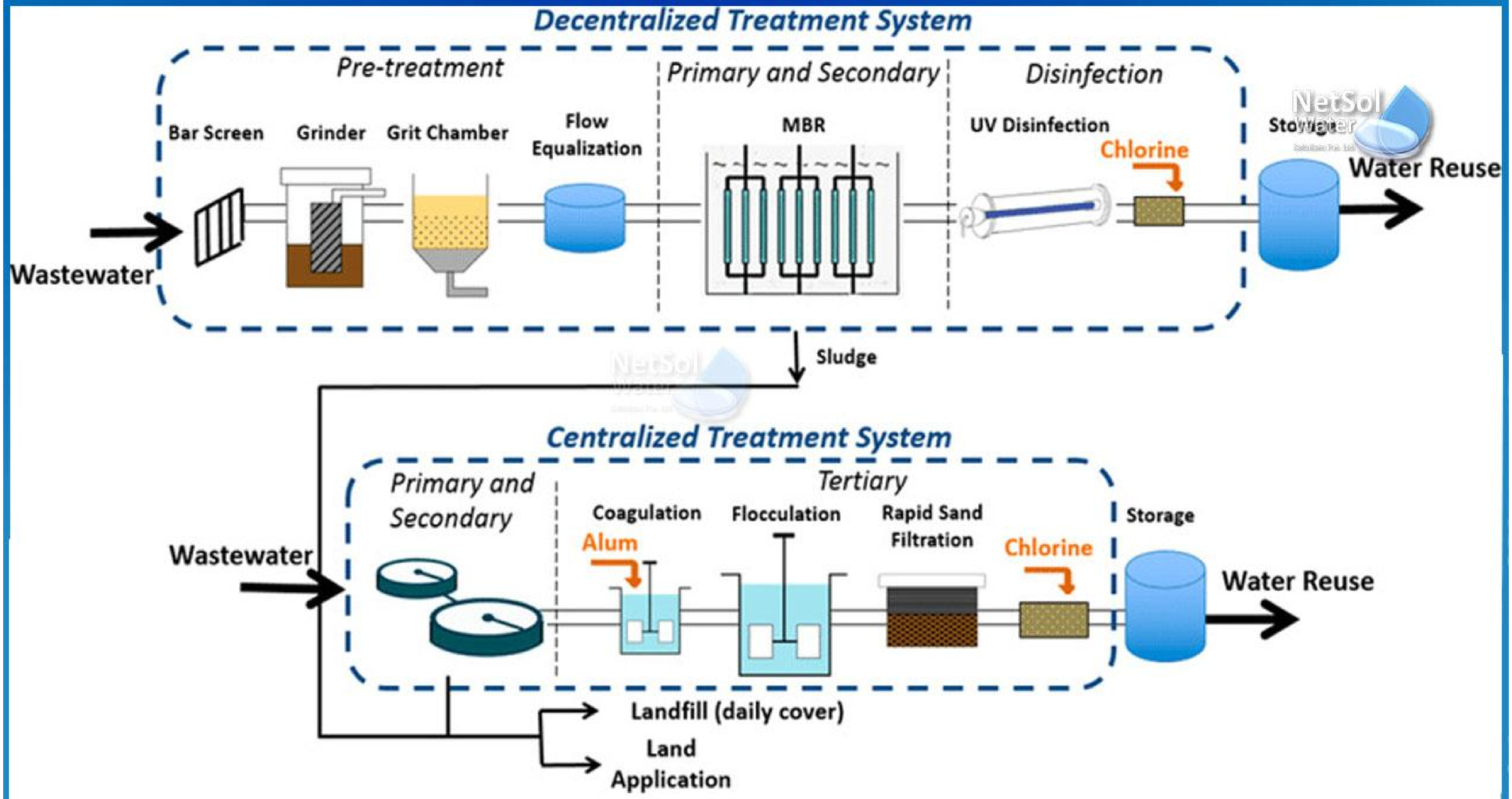
- Flows at any point in the system would remain small, implying less environmental damage from any mishap.
- System construction results in less environmental disturbances as smaller pipes would be installed at shallow depths and could be more flexibly routed.
- The system expansion is easier, new treatment centers can be added without routing ever more flows to existing centers.
- Entry of industrial waste could be more easily monitored.
- Sector wise treatment permits sewage transmission over shorter distances.
- Treatment units are close knit and are free from odours and insects.
- Lesser investment is required for the sewer pipelines.
- Community participation is essential; hence people can participate in the monitoring of the system performance. This instills confidence among the people.
- Quality of treatment is more efficient than traditional system due to accurate estimation of wastewater generation and lower quantity of wastewater;

# Disadvantages of Decentralized Wastewater Management Systems (DWWMs)

- Policies regarding installation, operation and maintenance are not yet well established in many of the developing countries.
- Standardization of the systems is difficult as significant variation exists with regard to technical design to suit the local geography and climatic conditions.
- Local people will have to bear all by themselves the O&M of the treatment plant.
- Getting a site for the STP may be difficult amidst built up sections and eventually, only the graveyards or cemeteries have to be the site.

# Situation Suitable for Decentralized Wastewater Management Systems (DWWMs)

- Where clusters of on-site systems are existing and there is no control on the fate of the pollutants
- Improper maintenance of on-site treatment systems and exorbitant cost of conventional remediation by implementation of centralized systems
- Community / institutional facility is far away from the existing centralized system
- Localities where there is scarcity of freshwater
- Localities where there is a possibility for localized reuse of treated wastewater
- Localities where discharge of partially treated wastewater is prohibited due to various environmental reasons
- Localities where extension of existing centralized system is impossible
- Newly developed or existing clusters of residences, industrial parks, public facilities, commercial establishments and institutional facilities




# Decentralized vs Centralized Water Treatment Systems

# COMPARISON BETWEEN CENTRALIZED & DECENTRALIZED TREATMENT

| CENTRALIZED | DECENTRALIZED |
|---|---|
| • Large-scale plants that serve large municipal or regional service regions characterize centralized treatment.<br><br>• The populace is provided with drinking water through huge, **centralized water treatment plants** in this strategy. The purified water is conveyed to all of the settlements in the treatment plant's service area, necessitating a large pipe network to reach even the most isolated communities. | • Smaller facilities are positioned close to the water source or treatment demand, serving a more localised area in decentralized treatment.<br><br>• **Decentralized treatment** refers to the practice of treating water or wastewater at the point of supply, demand, or, in the best-case scenario, both. |

# Natural methods of wastewater treatment

# WETLAND



- Wetlands are areas where water is either present at or near the surface of the soil for a significant part of the year. They support aquatic plants and have unique hydric soils.
- Wetlands are typically associated with decentralized wastewater treatment systems.

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# TYPES OF WETLAND



- Wetlands as a part of various decentralized approaches , includes :

  - ❑Natural Wetlands
  - ❑Constructed wetlands

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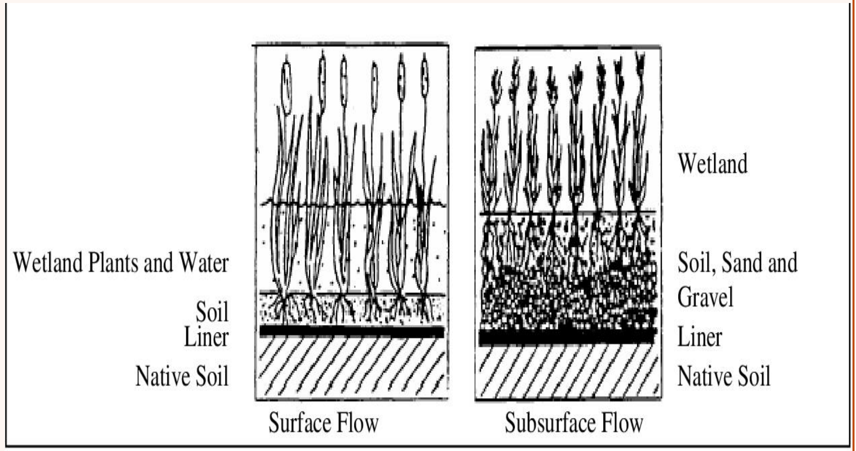
➢ Natural Wetland

These are ecosystems that naturally form in areas with water-saturated soils, such as swamps and marshes.

# CONSTRUCTED WETLAND



- Constructed wetlands are engineered systems designed to mimic natural wetlands for the purpose of wastewater treatment and ecological restoration.
- They utilize natural processes involving vegetation, soil, and microorganisms to remove pollutants from the water.
- They include :
  - ❑ Surface Flow Wetlands
  - ❑ Subsurface Flow Wetlands

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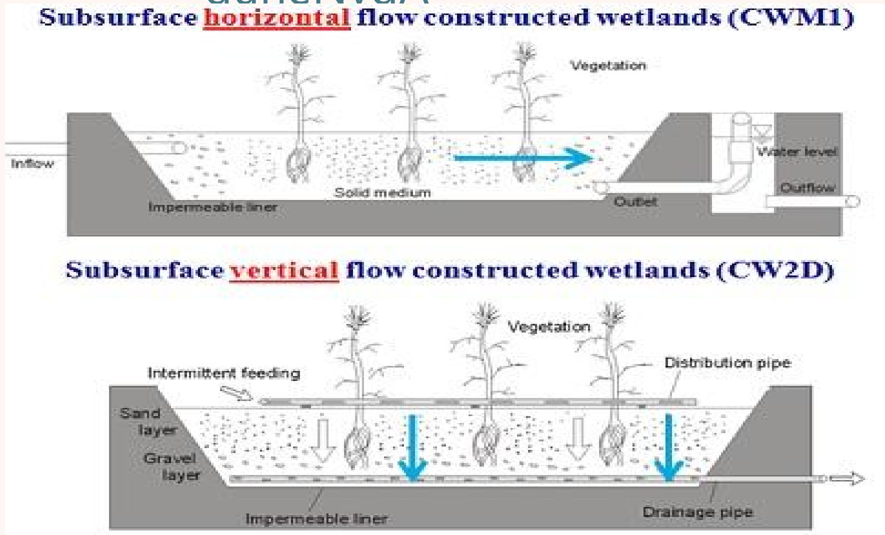
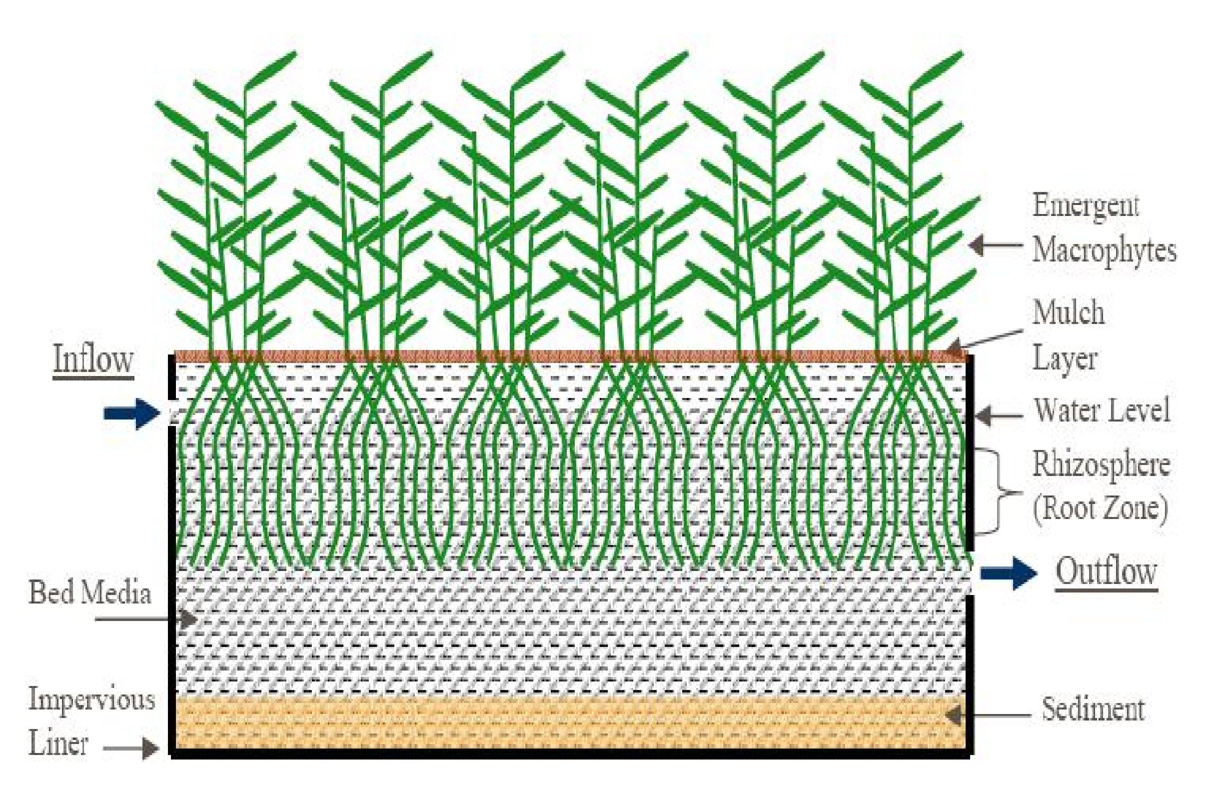


# COMPONENTS OF CONSTRUCTED WETLAND



- Inlet Structure
- Vegetation
- Substrate
- Water Flow System
- Outlet Structure
- Plant Root Zone
- Sediment Storage Area
- Maintenance Access
- Additional Features

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# PROCESS INVOLVED



- Inflow and Pre-treatment

It involves screening , settling and equalization.

- Primary treatment

This treatment mainly involves sedimentation.

- Secondary treatment

This occurs in surface flow as well as subsurface flow where the wastewater flow through soil surface , gravel bed ,vegetation allowing all physical , chemical , biological process resulting a surface for microbial growth.

# PROCESS INVOLVED (CONTD)



- Tertiary treatment

  Under this treatments comes filters and disinfection process.

- Effluent Discharge or Reuse

  The treated water is then discharged into the river or any other water bodies. It can also be reused for purposes like irrigation, industrial process.

# Lagoons

# Lagoons

- Cost-effective and energy-efficient solution for wastewater management
- **Aerated lagoons** use mechanical aeration to introduce oxygen into the water, promoting the growth of aerobic bacteria that decompose organic waste.
- **Anaerobic lagoons** function without the presence of air, creating conditions conducive for anaerobic microorganisms to thrive. These lagoons are typically used for high-strength waste streams.

# Aerated Lagoons

**Facultative aerated lagoons**

- Simple
- Low rate of aeration
- Upper aerobic, lower anaerobic
- Solids may settle down

**Aerobic aerated lagoons**

- Fully aerobic from top to bottom
- No settlement of solids
- If the effluent is settled and the sludge recycled -

# Aerated Lagoons

## Advantages

- Resistant to organic and hydraulic shock loads
- High reduction of BOD and pathogens
- No real problems with insects or odours if designed and maintained correctly
- Can treat high loads
- Less land required than for simple pond systems
- Treated water can be reused or discharged

## Disadvantages

- Requires a large land area
- High energy consumption, a constant source of electricity is required
- High capital and operating costs depending on the price of land and of electricity
- Requires operation and maintenance by skilled personnel
- Not all parts and materials may be locally available
- Requires expert design and construction and supervision

# Assignment -4

# PHYTORID –
# WASTE WATER TREATMENT TECHNOLOGY -
# CSIR NEERI